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# Practical AI for Academics

Day 2 — hands-on with your own materials

**Claes Bäckman**

Leibniz Institute for Financial Research SAFE

Three messages: ground the agent in *your* files, run it as a critic before you publish, and keep verification close.

### 1 From mental model to your draft

Day 1 was tools and theory — **today we point them at your own paper, prose, and slides.**

*Implication: the leverage is in materials only you have.*

### 2 Critic, not co-author

Eight subagents reading one draft beat one diffuse pass. The skill is **cheap pre-flight checking**, not replacing the referee.

*Implication: use AI to find what you would miss.*

### 3 The bottleneck is judgement

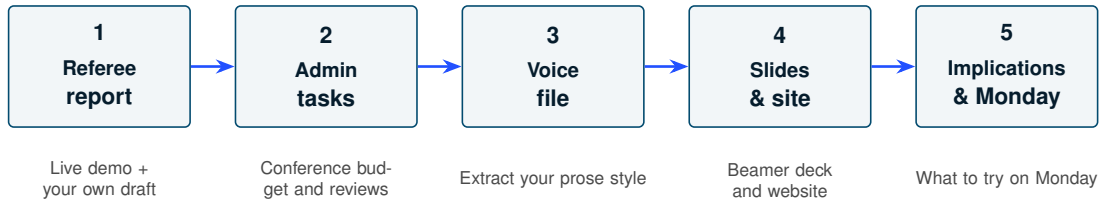
Producing draft prose, slides, and websites is now cheap. **Reading, checking, and disagreeing** is where the researcher's edge sits.

*Implication: verification is the skill to invest in.*

**Key takeaway.** By the end of today you will have run a referee report on your own draft, built a `voice.md`, and seen slides and a website generated from your materials.

## Day 2 agenda — five hands-on blocks, ending with implications.

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**What you walk out with.** A referee report on a draft you actually care about, a `voice.md` from your own writing, and a list of three things to try next week.

*Everything today assumes the Day 1 setup: VS Code, Claude Code, and a `CLAUDE.md` in the project folder.*

**Slides, skills, and guides for both days:** [claesbackman.com/aarhus](https://claesbackman.com/aarhus)

# A referee report on your paper

*Pick one draft you would like another pair of eyes on. Preferably a relatively close-to-finished draft*

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## How to get started

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Get the skills from [claesbackman.com/aarhus](https://claesbackman.com/aarhus) — download the zip, or paste the one-line terminal command on that page. Source and updates:  
[github.com/claesbackman/AI-research-feedback](https://github.com/claesbackman/AI-research-feedback).

Unzip, then copy each skill folder (review-paper, review-paper-light, ...) into  
~/.claude/skills/ — a hidden folder under your user folder (Windows:  
C:\Users\you\.claude\skills)

- To show hidden folders: Open File Explorer, click the View tab at the top, select Show, and check Hidden items
- A skill in ~/.claude/skills/ works in every project; one in project/.claude/skills/ only in that project

Restart VS Code

The skill loads when you type /skill-name

## Live demo — /review-paper on a real draft, eight subagents in parallel.

### Setup

- One of my own papers, in its own folder
- .tex source, .bib, supporting materials
- One command: /review-paper with a target journal (AER)

### Output

- A structured markdown report
- Referee-style comments, grouped by theme
- Specific line and section references

#### Eight subagents running in parallel

- Spelling, grammar & style
- Internal consistency & cross-references
- Unsupported claims & identification
- Mathematics, equations & notation
- Tables, figures & documentation
- Referee assessment: positioning & fit
- Contribution advocate
- Contribution skeptic

*Pattern: eight narrow critics beat one diffuse one.*

**Key takeaway.** The skill is a ~200-line markdown file. You can read it, edit it, and download it yourself.

## Hands-on — run a referee report on your own writing.

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1. Open one of your own drafts in VS Code
2. Make sure it is in markdown or LaTeX — convert if needed
3. Run `/review-paper-light` for a 1–2 minute version, or `/review-paper` for the full eight-agent run
4. Read the output. It's not all correct!

### Debrief questions

- What did the report catch that you would not have seen by yourself?
- What did it get wrong?
- Which subagent gave you the most useful feedback?



**A skill is a markdown file the agent loads on demand — invoke with `/name`.**

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A skill lives at `.claude/skills/your-skill/SKILL.md` (or `~/.claude/skills/` for every project):

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`name: review-paper`

`description: Run an 8-agent pre-submission referee report for an academic paper  
targeting a specified journal`

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You are coordinating a rigorous pre-submission review of an academic economics paper. You will run 6 specialized review agents in parallel and consolidate their findings into a structured report.

Invoke with `/review-paper`. Claude loads the instructions and runs them.

## CLAUDE.md is always loaded, skills are on demand.

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### CLAUDE.md — always loaded

- Loaded into context every session
- Background, conventions, voice
- “How I want you to work with me”

**Cost:** *permanent context budget.*

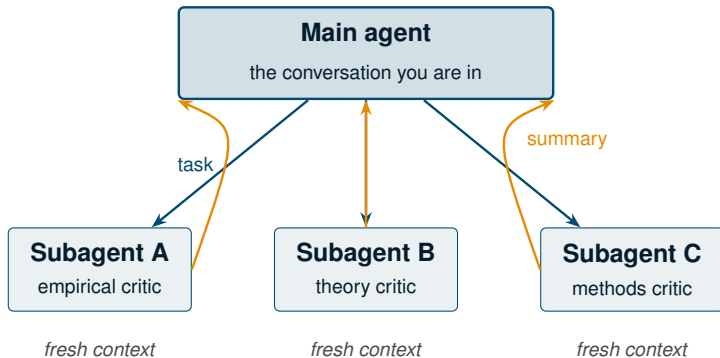
### Skill — on demand

- Loaded only when invoked with /name
- A specific workflow, with structured output
- “Do this thing for me”

**Cost:** *zero until triggered.*

**Key takeaway.** If you find yourself pasting the same prompt multiple times, you can ask Claude to make it into a skill.

**Subagents add two things the main agent cannot do alone — parallelism and context isolation.**



**Key takeaway.** The /review-paper skill spawns **eight** subagents.

## The second-opinion pattern — spawn an adversary to check on the work.

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*The main agent sees its own prior answer and has reason to defend it.*

### Why a fresh subagent is different

- No chat history: no anchoring on what you already wrote
- No conversational momentum: no reason to agree with the main agent
- Clean context: can hold the whole artifact at once

### When to spawn one

- Before accepting a long answer: *“is this actually right?”*
- After a long refactor: *“what did we break?”*
- When you suspect flattery: *“give me the honest version”*
- Before sending: *“read this as a hostile reader”*
- After doing automatic edits: *“look over the changes we did”*

**Key takeaway.** The phrases “**spawn a subagent**” or “**launch an agent**” are the trigger words. The main agent knows what to do and will write the instructions for you.

## What the demo is meant to show

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1. **Grounded output.** The report is about *your* draft, not a generic checklist
2. **Subagents are why this works.** Eight narrow critics catch things one diffuse reader misses
3. **Not a replacement for a referee.** It is a pre-flight check before you submit
4. **Transparent.** The skill is a  $\sim$ 200-line markdown file — fork it, modify it, make it yours

**Key takeaway.** The point is not to trust the report. The point is to make hard feedback cheap enough that you ask for it before submitting — not after the desk reject.

## Other tools generate referee reports too

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| Tool                             | What it does                         | Pricing       |
|----------------------------------|--------------------------------------|---------------|
| <b>Refine.ink</b>                | Referee-style report on a paper PDF  | <i>Paid</i>   |
| <b>coarse.ink</b>                | Lightweight version of the same idea | <i>≈ free</i> |
| <b>Referee 2</b>                 | Paid referee-report tool             | <i>Paid</i>   |
| <b>Stanford Agentic Reviewer</b> | Open agentic reviewer                | <i>Free</i>   |

*Many more out there — new ones launch every few weeks.*

**Key takeaway.** The advantage of running your own skill is that it lives next to your draft, knows your conventions, and you can edit what it checks.

*Now launch another agent to critique the referee report you just generated — ask Claude to write the instructions.*

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# Working with secure data



## Secure servers split the workflow

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### What Claude Code can still help with on your laptop

- Your `.do` files
- Writing and iterating on empirical strategy

Claude cannot run the files or read the logs on the server, but can still be very helpful in a number of ways

## Two artifacts to keep local

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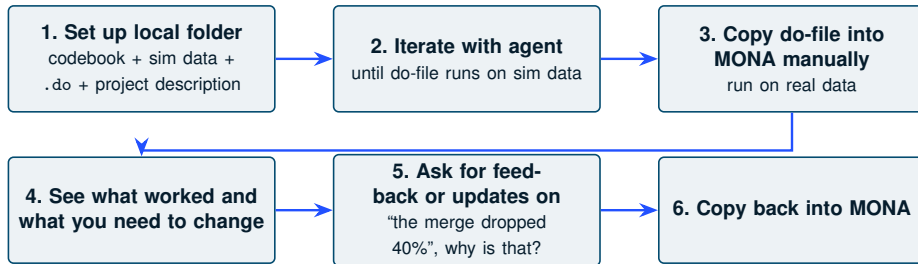
### 1 Codebook in markdown

- Convert once (`/pdf-to-markdown`)
- Variable names, labels, value ranges, missing-value codes
- Agent reads it before writing `recode`, `label` `define`, `merge`

### 2 Simulated dataset

- Same variable names, types, ranges — not real values
- AER/JF/ReStud replication packages often ship with these
- Or generate from the codebook

## A workflow for secure data on servers



**Key takeaway.** The agent never sees the real data — but it sees everything that explains the data. That is usually enough to write correct Stata code.

# Organising a conference

## Live demo: Administrative tasks Claude Code can take off your desk.

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### Setup

- A folder with a call for papers

### The goal

- Build a budget for the conference

### The output

- A budget spreadsheet with line items, totals, and assumptions

**What we are using.** Plain Claude Code prompts — no skills, no MCP. Just the folder and the agent.

## Two points the admin demo is meant to show

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### 1. AI tools reduce the cost of administrative work.

- Not just for research
- Often the highest-leverage use case, because the friction is the highest and the quality bar is the lowest

### 2. Context is the difference between useful and useless.

- A chat tool gives you a generic conference-budget template
- Claude Code finds your keynote speakers, your dinner, and your PhD travel funding by reading the folder

**Key takeaway.** The same pattern — open the folder, hand the agent the materials, ask for the output — works for grant reports, course design, referee assignments, and committee memos.

# Writing with agentic AI

*What are the three phrases you hate seeing in AI-edited prose?*

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## A `voice.md` file makes edits sound like you

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When editing my prose, match the voice in `voice.md`.  
Do not use em-dashes, bullet lists, or passive constructions unless the original does.  
Do not use "it's not x, it's y"-style writing.

### What goes in it

- 5–10k words of your prose — intros, abstracts, paragraphs you are proud of

**Key takeaway.** Reference it from `CLAUDE.md` to give Claude writing context for every session

## Hands-on (15 min) — build a `voice.md` from your own writing.

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We use a skill written by Mihail Velikov: `voice-extractor`.

1. Upload 4 PDFs of your own writing to a folder
2. Call the skill: `/voice-extractor sample-papers`
3. Read through `voice.md` and verify — edit anything that does not sound like you

**What goes in.** Intros, abstracts, paragraphs you are proud of. 5–10k words is plenty.

**What stays out.** Co-authored sections where the prose is not yours. AI-edited drafts.

**Key takeaway.** Reference `voice.md` from `CLAUDE.md`. Every future session loads your voice into context.

## How to tell whether your `voice.md` is working

### Signs it is working

- Edited paragraphs read more like you, less like a generic LLM
- Fewer em-dashes, fewer bullet lists, fewer “crucially” / “moreover”
- You stop having to say “preserve my voice” in every prompt
- Your coauthors stop noticing which paragraphs were AI-edited

### Signs it is not

- Output still sounds generic
- AI tics persist after editing
- You keep adding “don’t use X” to every prompt

***Fix:** sample is too short, or includes too much AI-edited prose. Add more of your writing.*

**Key takeaway.** `voice.md` is a living file. Update it when your style shifts, or when an editor pushes you in a direction you want to keep.

## AI detectors: treat the score as a signal, never as a verdict.

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**How they work.** A classifier trained on the fingerprint of model text — predictable word choice, even rhythm, stock phrasing. It returns a probability, not evidence.

**Why the first generation failed.** False positives, concentrated on non-native writers and formulaic academic prose. OpenAI withdrew its own classifier in 2023; several universities switched off Turnitin's score.

**Pangram.** The strongest current detector, with very low reported false-positive rates. Still a probability, and it cannot tell you *how* AI was used: a student who used it to learn, then wrote the essay, looks like one who never touched it.

### What to do instead

1. **Never accuse on a score alone.** Use it to decide whom to talk to, not what grade to give. Ask the student to explain their work.
2. **Set a per-assignment AI policy.** State plainly what is allowed where — ambiguity causes the problems, not the tool.
3. **Move the stakes to where understanding shows:** closed-book exams, in-class work, oral exams. Then using AI to learn pays off and avoiding learning does not — and detection matters much less.
4. **Teach the good use.** *“Act as a tutor: ask me questions and give hints, do not give the answer.”* Publish answer keys.

## SECTION 5

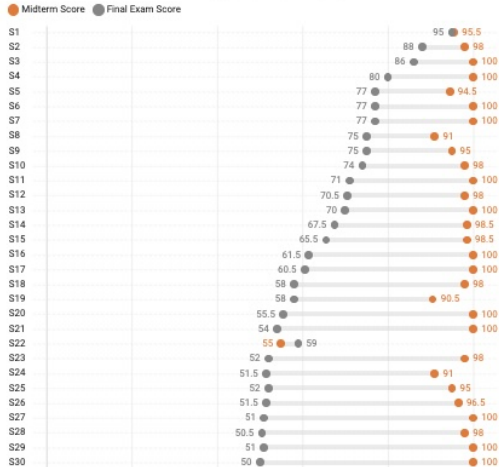
# Teaching

*Go through your syllabus. Which take-home assignment worth non-trivial credit could a student complete with a frontier model today?*

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## ECON 1170 Midterm & Final Scores

After Serrano made the final in-person, 18 students dropped the class and nine students did not take the exam. The scores of the remaining 59 students are displayed here.



## AI breaks the grade–knowledge link, and the pre-AI classroom is no benchmark.

### Three things AI changes, according to Bryan

1. **The grade–knowledge link is broken.** A good take-home grade no longer shows the student can do the thing.
2. **Students need to use AI well later.** Banning it everywhere leaves them untrained.
3. **AI should let us teach more.** Tutoring, feedback, and individual practice used to be too expensive.

*Bryan (2026). All Day TA, used in several of his examples, is Bryan's own company.*

#### The pre-AI status quo was already leaky

- US college study time fell from **40 to 27 hours a week**, 1961–2003
- A quarter of finance students copied **wrong** Chegg answers, and learned less

#### And AI makes homework a worse signal

- Post-AI: homework scores +18%, completion time –30%, exam performance –20% (China)
- Time on AI-susceptible homework collapsed. Exam performance on those topics fell 25%

**Key takeaway.** Modify the course, the expectations, and the evaluation, and education can come out **more** effective than before.



## Eight rules, two halves: protect AI-free learning, then use AI to teach more.

### Protect learning

1. **Decide what should be learned without AI.** Be explicit about which assignments test AI-independent competence.
2. **Teach better than the pre-AI status quo.** Going back a generation is not the goal.
3. **Give students an incentive to learn.** Regular, graded checks. Effort follows tests.
4. **Do not give AI-cheatable assignments.** Every take-home worth real credit can be done by a frontier model today.

### Teach more with AI

5. **Help students learn more efficiently.** Class-specific tutoring, spaced repetition, mastery learning at scale.
6. **Use AI to improve your own teaching.** Find out where you were confusing before the exam tells you.
7. **Personalise assignments.** Problems at the level each student is struggling with.
8. **Raise standards.** If research and drafting are cheaper, the A-grade bar moves up.

**Key takeaway.** Rules 1–4 are about the **assessment** you already have. Rules 5–8 are about things you could not afford before. Start with the first half.

## Rules 1 and 4: audit what each assignment certifies and whether AI can do it.

### How to run the audit

1. Syllabus, all assignments, one past exam in a folder
2. Ask Claude Code for the audit (right)
3. Make it *actually attempt* one assignment. Grade it with your rubric
4. Decide per component: keep, move in-class, redesign

### Two questions per component

- What is it a *proxy* for? Nobody inverts a matrix by hand, but being able to shows you understand the regression
- Is it needed *without* AI? A French essay written with AI is useless once translation is perfect

#### Prompt

*“Read the syllabus and every assignment in this folder. For each graded component, give me a table with: (1) the competency it is meant to certify, (2) whether students need that competency without AI, (3) whether a student using a frontier model could get a top mark on it today, and (4) what you would change. Be blunt. Then complete problem set 2 yourself, as a student would, and tell me what grade it would get.”*

*Expect the agent to score well. That is the point of the exercise.*

**Key takeaway.** Take-home work should no longer count, Bryan wrote on the day ChatGPT launched.

## Three ways to make an assignment un-cheatable, and what each one costs you.

### 1 In-class only

Oral exams (*viva voce*), blue-book written exams, in-class problem solving.

**Works because** AI is not in the room.

**Cost:** very time-intensive to evaluate. Hard to scale without TAs.

### 2 Use AI to fight AI

Students hand in a précis early. You show the class what a frontier model produces from it. The final paper must be *better than that*.

**Works because** the AI output becomes the floor. To beat it: archives, phone calls, site visits, own data.

**Cost:** you generate and read the baseline. Term papers only.

### 3 Un-cheatable at home

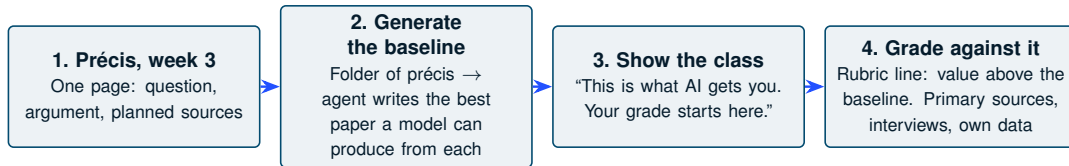
Low-stakes quizzes graded 0 or 1. Pass with 10 or more correct, *however many are wrong*. Wrong answers require the student to explain their logic.

**Works because** more right answers buy nothing, so cheating buys nothing. Explaining errors is where the learning happens.

**Cost:** question bank and automation up front. Scales to any size.

**Key takeaway.** Bryan runs option 3 via his product, All Day TA. The design works with any quiz tool.

## The précis workflow, step by step: make the model's paper the grade floor.



### Prompt for step 2

*“For each précis in `precis/`, write the strongest 2,000-word term paper a frontier model can produce from it, using only what it already knows. Save as `baseline/<student>.md`. Then add a one-paragraph note on what a student would need to do to clearly beat this paper.”*

### Practical notes

- Tell students exactly *how* to beat the baseline. Bryan’s list: get into the archives, get on the phone, visit the site, create data
- Detection is a backstop, not the design. Pangram’s false-positive rate is very low (Day 1), but the workflow works even if you never run it
- Require an attached note on what AI did (see the policy slide)

## Rule 3: students study when tested. Regular graded checks beat study plans.

### What the evidence says

- Study-tool use rises **5–10×** in exam weeks, and is far higher in courses with exams
- Study plans, reminders, and grade coaching **barely moved** performance (Oreopoulos et al.)
- An intermediate exam raised **pass rates and final grades**
- A goal for completing practice exams worked. A goal for the final grade did not

*Take-home work can no longer force effort, so the checks must be frequent, graded, and hard to outsource.*

### How to implement it

1. Weekly quiz, pass/fail, small weight each, most weeks count
2. Generate the bank from your own notes:  
*"From notes/week3.md, write 15 multiple-choice questions at three levels, with distractors built from common misconceptions and a one-sentence explanation each. Output in our LMS import format."*
3. Import (GIFT for Moodle, CSV for Brightspace). Randomise question order and selection
4. Verify every answer key yourself before it goes live

**Key takeaway.** The quiz is not the assessment. It is what makes the studying happen before it.

Rules 5–7 rest on trial evidence: AI tutoring, teacher feedback, adaptive practice.

| Idea                       | Evidence Bryan cites                          | Effect                    |
|----------------------------|-----------------------------------------------|---------------------------|
| AI tutor (rule 5)          | Six-week trial with Nigerian students         | 0.31 SD                   |
|                            | World Bank meta-analysis of older ed-tech     | 0.12 SD                   |
|                            | Khanmigo (NBER); Harvard physics AI tutor     | Gains despite limited use |
| Teacher feedback (rule 6)  | Feedback RCT, 1,000+ online CS teachers       | +13% use of student ideas |
| Adaptive practice (rule 7) | Mindspark adaptive system (pre-LLM), 4 months | 0.37 SD maths, 0.23 Hindi |
|                            | LLM-chosen problem sets (Taiwan), no-AI final | 0.15 SD vs fixed sequence |
| Fetching answers from AI   | Homework up, exams down (previous slides)     | Does not work             |

**Key takeaway.** Old mechanisms, new cost: tutoring, mastery learning, spaced practice at scale.

## Implementing rules 5–7 for free: tutor, misconception digest, levelled problem sets.

### Course tutor

A shared assistant (custom GPT, Claude project, NotebookLM) loaded with *your* lecture notes and problem sets.

**System prompt essentials:** use the course's notation, never give the answer, ask the student to explain first, return to topics they missed.

*Grounding and instructions matter more than the tool.*

### Misconception digest

Each week, export the questions students asked (tutor logs, forum, email), anonymised.

**Prompt:** *“Group these by underlying misconception, rank by frequency, and point to the lecture slide most likely to have caused each.”*

*Fix the slide before the next lecture, not after the exam.*

### Levelled problem sets

One problem, three difficulty tiers. Assign the tier by last week's quiz result.

**Prompt:** *“Write this problem at three levels: scaffolded, standard, extension. Same concept, full solutions, and a rubric I can grade against.”*

*The agent drafts and pre-grades. You spot-check and decide.*

**Key takeaway.** Student questions are personal data. Anonymise before exporting and check the institutional policy once.

## Rule 8: raise the bar, and write the AI policy into the syllabus.

### Bryan's course policy, verbatim

*"You may use AI in this course. If you use AI to write, you must specify in an attached note precisely what AI assisted with. Understand that with AI, my expectation for the quality of your ideas and writing is higher than it used to be — you need to show that you are able to provide value above and beyond AI. For readings, you must read them first yourself! Do not summarize them with AI. It will be very obvious in class if you do. I want your ideas; I can ask GPT for its thoughts without coming to class, as can your classmates."*

**Key takeaway.** The clearer you are about where students **should** use AI and where they **must not**, the better they get the balance right.

### Why raise the bar

- Randomised workplace studies: AI raised writing quality and cut completion time
- Consultants working inside AI's competence produced better work ([Dell'Acqua et al., 2026](#))
- If content-gathering is cheap, an A from ten years ago is no longer an A

### Practical

- State the policy *per assignment*, not once per course
- Require the attached AI-use note and grade it
- Rubric line: sloppy citation and error-ridden prose now cost more, not less



## Hands-on (15 min) — audit one of your own courses.

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1. Put the syllabus and the graded assignments for one course in a folder. One past exam if you have it
2. Open the folder in VS Code and run the audit prompt from the Rules 1 and 4 slide
3. Read the table. Where do you disagree with the agent about what a component certifies?
4. Ask it to attempt the assignment you are most worried about. Grade the result with your rubric
5. Pick one component and redesign it using one of the three options: in-class, précis baseline, or threshold quiz

### Debrief questions

- Which assignment surprised you most?
- Which of Bryan's eight rules is cheapest to implement in your course next semester?
- Which one would your department push back on, and why?

# **Slides and websites**

# Making slides with AI

## The old path

- Open Beamer
- Write LaTeX by hand
- Fight the syntax
- Iterate slowly

*Hours of mechanical work between idea and first draft.*

## The Claude Code path

- Open the project folder
- Hand Claude a file: *"Make a Beamer deck based on @paper.tex"*
- Compile, look at the output, iterate

*This deck was built that way.*

**Key takeaway.** This also means that we have to practice presentations in a different way

## Including a style file gives Claude guidance to match the look you want.

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**Without a style file.** Claude defaults to generic Beamer themes — bullet-heavy, blue-and-white, no action titles.

**With a style file.** The deck inherits your colour palette, frame structure, and title conventions.

### How to build a `slides-style.md`

- Upload a few old presentations
- Ask Claude to analyse the structure, palette, and title style
- Save the result as `slides-style.md` in the project folder
- Reference it whenever you ask for a new deck

**Key takeaway.** The Day 1 deck, this deck, and your future decks can all share one style file. Build it once.

## Live demo — transcript and code → interactive website.

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### The task

- Take a folder of do-files and a transcript
- Build a standalone interactive website

### What Claude Code does

1. Reads the code and the transcript
2. Sees how the code constructs the figures
3. Writes a standalone HTML page — one file, no dependencies

#### Why this works

- Claude knows how the figure was built — it has the code
- Claude knows how to write HTML
- Putting them together is exactly where agentic AI has an advantage over chat

***Pattern:** two skills the agent already has, glued together by the folder.*

**Key takeaway.** Same pattern works for policy briefs, replication packages, course websites, and conference programmes.

## SECTION 7

# Literature

Convert files,

## Zotero MCP connector

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**MCP servers plug the agent into external systems — know it exists, install only with a clear use case.**

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**Model Context Protocol.** A standard for letting Claude call external tools beyond the local filesystem.

### **Examples already in production**

- GitHub — open PRs, read issues, comment on code review
- Databases — query Postgres, BigQuery, etc. directly
- Custom servers you wrote yourself

*Like browser extensions, but for the agent. The list is growing fast.*

**Key takeaway.** For most academics, today, **you do not need MCP**. Know the concept exists; install one only when you have a concrete use case.

# Implications and getting started

## Separate publishing from research quality

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### Publishing gets noisier

- Journals fill with polished, competent-looking mediocrity
- Signal-to-noise drops and reviewing gets harder
- Editors and referees absorb the cost

### Research can still get better

- Ideas that could not be written before reach readers
- The language barrier falls
- Feedback, formerly a privilege of elite departments, becomes universal

**Key takeaway.** The open question is whether we update how we **evaluate** research now that competent-looking writing is free. Because research is about to get better.

## What this means for academic work

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- **Friction drops an order of magnitude** in writing, editing, and admin
- **The bottleneck shifts** from producing to judging
- **Verification becomes the differentiator** — not typing speed, not LaTeX fluency
- **Disclosure norms are forming** — expect them to harden over the next year
- **Tooling will keep moving** — the principles (specificity, iteration, verification) will not

**Key takeaway.** The researchers who get the most out of these tools are not the most technical. They are the ones who treat AI as a colleague to argue with.

## Verification is going to be increasingly important (and in the end also automated?)

| Check               | What to look for                                                     | How to do it                                 |
|---------------------|----------------------------------------------------------------------|----------------------------------------------|
| Run the code        | Claude writes plausible Stata / R / Python that does not always run. | Run the code yourself.                       |
| Check the numbers   | Statistics, dates, quotes get invented.                              | Re-derive from the table or paper.           |
| Read the citations  | Citations to papers that do not exist are common.                    | Check if you can find it on Google Scholar.  |
| Re-read the passage | Claude confidently describes text not in the draft.                  | Open the file, read the section in question. |

**Key takeaway.** You are faster with AI, not absent.

## Disclosure and data norms are still in flux.

### Disclosure

- **Journals** mostly require acknowledging AI assistance but specifics vary
- **Policy briefs, teaching, internal docs** have no strong norm yet (?)
- **Default rule:** disclose where your name is on it

### Data handling

- Agentic tools run *locally*, but prompts and file excerpts may be sent to the model provider
- **Do not** paste confidential microdata, HR-sensitive material, or DUA-covered material into any AI tool without checking institutional policy
- Web chat  $\neq$  local agent — different tools, different policies

**Key takeaway.** Check before pasting. Once is enough — the policy almost never changes mid-project.

## Three things to try on Monday — and three habits to keep.

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### Three things to try Monday

1. Open your current project in VS Code. Run `/init` to draft `CLAUDE.md`, then edit it.
2. Run `/review-paper-light` on a draft you are stuck on. Read the output critically.
3. Build a `voice.md` from 5–10k words of your own writing. Reference it from `CLAUDE.md`.

### Three habits to keep

- **Always run, re-read, verify.**
- **Preserve your voice.** Push back when the edit sounds generic.
- **Ask for critique, not flattery.** “Be harsher” is a valid instruction.

**Key takeaway.** A one-hour Monday investment in `CLAUDE.md` and `voice.md` compounds across every later session.

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# Q & A

*Open floor. Email: [claes.backman@gmail.com](mailto:claes.backman@gmail.com) |  
Slides, skills, guides: [claesbackman.com/aarhus](https://claesbackman.com/aarhus)*



## References I

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Kevin A. Bryan. Eight rules for teaching in AI world. <https://kevinbryanecon.com/teachwithai.html>, August 2026.

Fabrizio Dell'Acqua, Edward McFowland III, Ethan Mollick, Hila Lifshitz, Katherine C Kellogg, Saran Rajendran, Lisa Kraye, François Candelon, and Karim R Lakhani. Navigating the jagged technological frontier: Field experimental evidence of the effects of artificial intelligence on knowledge worker productivity and quality. *Organization Science*, 37(2):403–423, 2026.

## Appendix — CLAUDE.md starter template.

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# About me

I am a researcher in [field].

# How I want you to work with me

- Ask clarifying questions before generating long output.
- Critical, skeptical tone in feedback. Do not flatter.
- When editing, preserve my voice (see voice.md). No generic LLM phrasing.
- Avoid bullet lists and passive voice in formal writing.
- Cite the specific line or section you are commenting on.

# Things to avoid

- Do not fabricate citations.
- Do not over-claim causality.
- Do not insert emoji or markdown decorations in formal docs.

## Appendix — instructor's skill library.

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| Skill                            | Use                                                |
|----------------------------------|----------------------------------------------------|
| <code>/review-paper</code>       | Pre-submission referee report, 8 agents            |
| <code>/review-paper-light</code> | Fast 2-agent version, $\approx$ 1 minute           |
| <code>/review-paper-code</code>  | Paper + code alignment review                      |
| <code>/review-grant</code>       | Pre-submission grant review, 6 agents              |
| <code>/review-pap</code>         | Pre-analysis-plan review                           |
| <code>/audit-analysis</code>     | Adversarial audit of changed analysis code         |
| <code>/explain-diff</code>       | Explain a code change as an HTML page with a quiz  |
| <code>/paper-version</code>      | Paper $\rightarrow$ policy brief / 1-page / 5-page |
| <code>/pdf-to-markdown</code>    | Convert PDFs to readable markdown                  |

All at [github.com/claesbackman/Al-research-feedback](https://github.com/claesbackman/Al-research-feedback) (MIT), linked from [claesbackman.com/aarhus](https://claesbackman.com/aarhus).  
Fork them, modify them, make them yours.

## Appendix — file format quick reference.

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| Format                                | Handling                                          |
|---------------------------------------|---------------------------------------------------|
| .md, .tex, .txt, .bib                 | Native — no conversion                            |
| .py, .R, .do, .ipynb                  | Native — read, edit, run                          |
| .csv                                  | Native up to a few thousand rows                  |
| .docx                                 | Convert: <code>pandoc file.docx -o file.md</code> |
| .xlsx                                 | Export to CSV first                               |
| .dta, .rds                            | Script-based conversion to CSV                    |
| <b>PDF (born-digital, short)</b>      | Native; expect table mangling                     |
| <b>PDF (long, scanned, equations)</b> | Convert to markdown first                         |

## Appendix — signs of session drift.

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- Claude ignores CLAUDE.md
- Repeats mistakes you already corrected
- Invents files that do not exist
- Gives circular answers
- Forgets the last three turns

**Fix.** Start a fresh session or close and reopen. Re-state the task. Re-attach the relevant file.

*30+ turns or the task has shifted → start fresh. Always.*